

SOFTWARE CONSTRUCTION AND DEVELOPMENT



TODAY'S AGENDA

- Introduction to the Course
- Objectives of the course
- Basic Concepts

COURSE INTRODUCTION

- **Course Title:** Software Construction & Dev
- **Total Credits:** 2+1 Credits
- **Pre-requisite:** Software Design & Architecture

CLASSROOM CODE

w5asxnd

COURSE LEARNING OUTCOMES

- **CLO 1:** Understand the bad and good code practices
- **CLO 2:** Understand the role of design and its major activities within the OO software development process, with focus on the Unified process
- **CLO 3:** Perform the dynamic modeling for real time problems

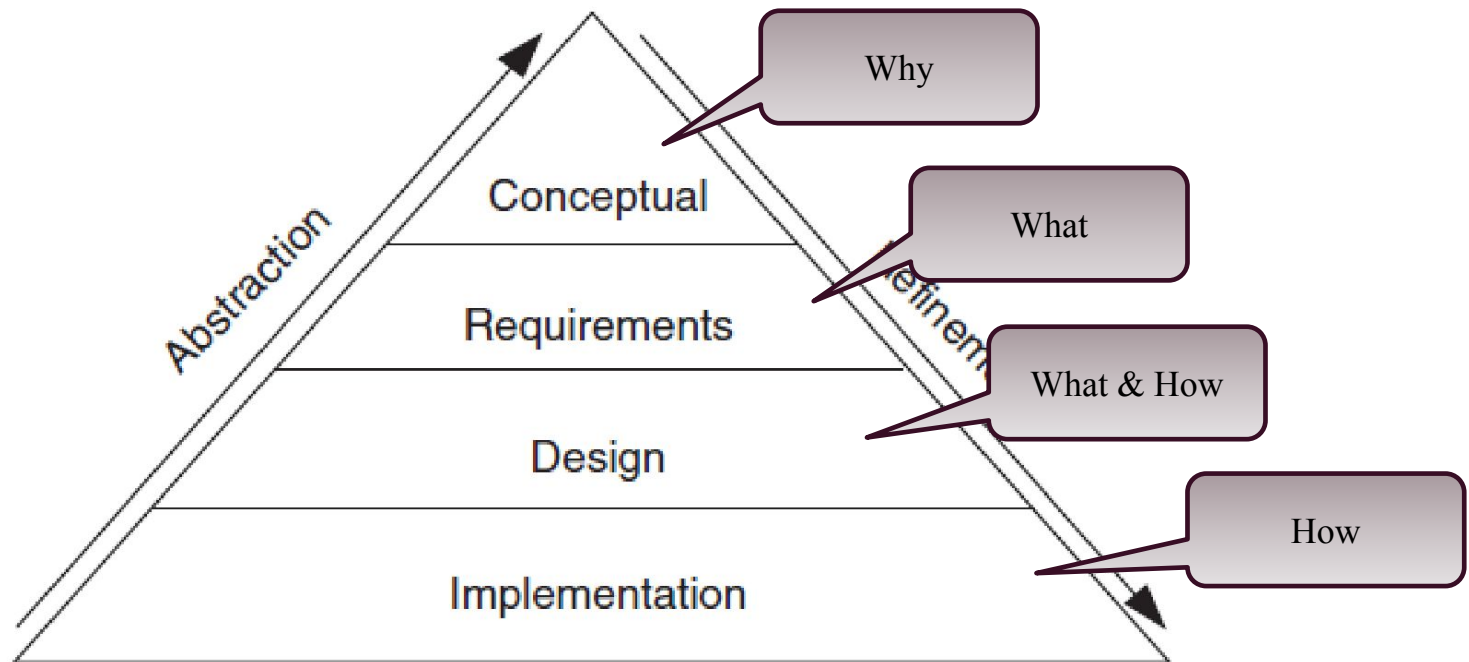
RECOMMENDED BOOKS

- Code Complete 2nd edition: A practical handbook of software construction, published by Microsoft Press, 2004
- Clean Code by Robert C Martin
- Software Engineering by Ian Sommerville

GRADING SCHEME

- Quizzes-10%
- Assignments -10%
- Presentation-5%
- Mids-25%
- Finals-50%

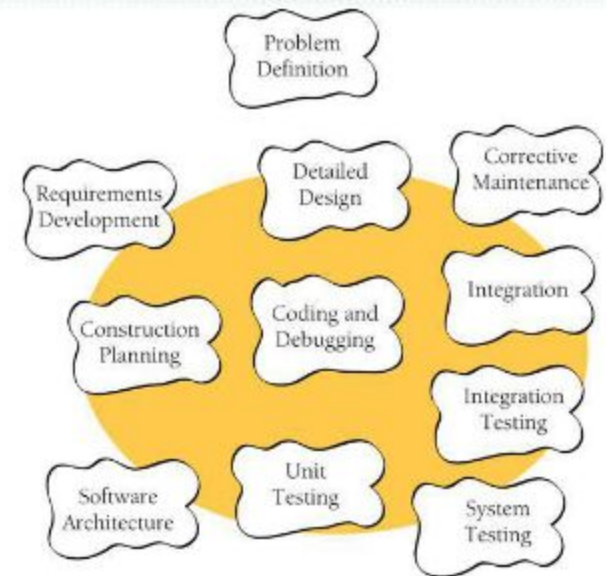
SOFTWARE DEVELOPMENT PHASES



SOFTWARE DEVELOPMENT

Developing computer software can be a complicated process, and in the last 25 years, researchers have identified numerous distinct activities that go into software development. They include

- Problem definition
- Requirements development
- Construction planning
- Software architecture, or high-level design
- Detailed design
- Coding and debugging
- Unit testing
- Integration
- System testing
- Maintenance



SOFTWARE CONSTRUCTION ACTIVITIES

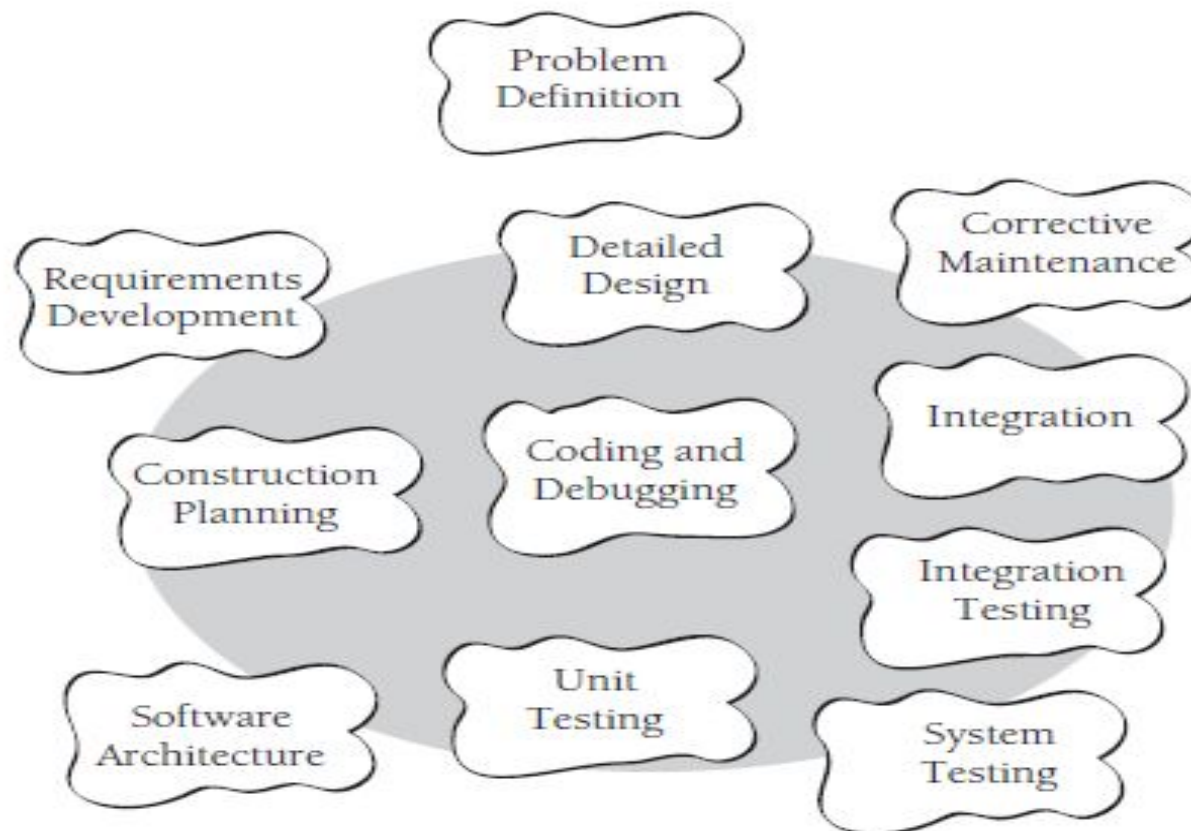


Figure 1-1 Construction activities are shown inside the gray circle. Construction focuses on coding and debugging but also includes detailed design, unit testing, integration testing, and other activities.

SOFTWARE CONSTRUCTION

- As the figure indicates, *construction is mostly coding and debugging* but also involves detailed design, construction planning, unit testing, integration, integration testing, and other activities.

SOFTWARE CONSTRUCTION

Here are some of the specific tasks involved in construction:

Verifying that the groundwork has been laid so that construction can proceed successfully

- Determining how your code will be tested
- Designing and writing classes and routines
- Creating and naming variables and named constants
- Selecting control structures and organizing blocks of statements
- Unit testing, integration testing, and debugging your own code
- Reviewing other team members' low-level designs and code and having them review yours
- Polishing code by carefully formatting and commenting it
- Integrating software components that were created separately
- Tuning code to make it faster and use fewer resources

WHY SOFTWARE CONSTRUCTION IS IMPORTANT

- Construction is a large part of software development
- Construction is the central activity in software development
- With a focus on construction, the individual programmer's productivity can improve enormously
- Construction's product, the source code, is often the only accurate description of the software
- Construction is the only activity that's guaranteed to be done

INTRODUCTION

- The term software construction refers to the detailed creation of working software through a combination of coding, verification, unit testing, integration testing, and debugging.
- Quality (or lack thereof) is very evident in the construction products
- Construction highly related to Computer Science due to
 - Use of algorithms
 - Detailed coding practices

SOFTWARE CONSTRUCTION FUNDAMENTALS

- Minimizing complexity
- Anticipating change
- Constructing for verification
- Reuse
- Standards in software construction

MINIMIZING COMPLEXITY

- Humans are severely limited in our ability to hold complex information in our working memories
- As a result, minimizing complexity is one the of strongest drivers in software construction
- Need to reduce complexity throughout the lifecycle
- As functionality increases, so does complexity

MINIMIZING COMPLEXITY

- Accomplished through use of standards
- Examples:
 - UML for modeling all aspects of complex systems
 - High-order programming languages such as C++ and Java
 - Source code formatting rules to aid readability

ANTICIPATING CHANGE

- Software changes over time
- Anticipation of change affect how software is constructed
- This can effect
 - Handling of errors
 - Source code organization
 - Code documentation
 - Coding standards

CONSTRUCTING FOR VERIFICATION

- Construct software that allows bugs to be easily found and fixed
- Examples:
 - Enforce coding standards
 - Helps support code reviews
 - Unit testing
 - Organizing code to support automated testing
 - Restricted use of complex or hard-to-understand language structures

REUSE

- Reuse refers to using existing assets in solving different problems.
- In software construction, typical assets that are reused include libraries, modules, components, source code..
- Reuse is best practiced systematically, according to a well-defined, repeatable process.
- Systematic reuse can enable significant software productivity, quality, and cost improvements.

STANDARDS IN CONSTRUCTION

Standards which directly affect construction issues include:

- Programming languages
 - E.g. standards for languages like Java and C++
- Communication methods
 - E.g. standards for document formats and contents
- Platforms
- Tools
 - E.g. diagrammatic standards for notations like the Unified Modeling Language

KEY CONSTRUCTION DECISIONS

Every programming language has strengths and weaknesses. Be aware of the specific strengths and weaknesses of the language you're using.

Establish programming conventions before you begin programming. It's nearly impossible to change code to match them later.

More construction practices exist than you can use on any single project. Consciously choose the practices that are best suited to your project.

Ask yourself whether the programming practices you're using are a response to the programming language you're using or controlled by it. Remember to program *into* the language, rather than programming *in* it.

Your position on the technology wave determines what approaches will be effective—or even possible. Identify where you are on the technology wave, and adjust your plans and expectations accordingly.

HOME TASK

- Exploring any recent research article related to the software construction and summarize in your own words.